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# AMBROSE ALLI UNIVERSITY, EKPOMA

CSC 309 — Past Questions & Materials

**by Renance**

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Course: CSC 309

Institution: Ambrose Alli University, Ekpoma

Papers in this pack: 1

Levels: 300

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## Paper 1 of 1 — CSC 309

Level 300 • First Semester • PastQuestion

AMBROSE ALLI UNIVERSITY, EKPOMA

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

CSC 309-TEST : DISCRETE MATHEMATICS

INSTRUCTION: ANSWER Q1 AND ANY OTHER ONE. TIME: 1 HOUR

(alla. Define the following terms: ech (ii) a complete graph (iii) a directed graph

A b. A football tournament consists of 5 teams:  $F_1, F_2, F_3, F_4$ , and  $F_5$ . Assume that after a number of games have been played we have the following situation:

'  $F_1$  has played  $F_2, F_3, F_4$  and  $F_5$  c. Consider the digraph :

$F_1$  has played  $F_2, F_3$ , and  $F_5$

$F_3$  has played  $F_4, F_2$ , and  $F_5$

$F_4$  has not played any team

$F_5$

aoe

$F_5$  has played  $F_4$ , "ADeterniine the adjacency matrix

cA Find the square of the adjacency matrix.

You are required to represent the above situation by a graph and write out the vertices.

2a. Define the fa!

H nath fit 1

ae. what is the difference between the matrix of a ca and its adjacency?

Ali). Determine whether the digraph with given adjacency matrix below is strongly connected.

r eh

UN o1i1i1 ,

&}/0011

$A = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$

Hhoo1 0 i Determide tho dgie oF matics, if any.

Q3a. A sample of 80 car owners revealed that 24 owned jeep and 62 owned. cars which are not je

Find the number of people who owned both jeep and a some other car.,

b. Draw the venn diagram representing the above question.

% ; ; 35 Uy

c.  $A = \{1, 3, 5, 7\}$  and  $B = \{2, 3, 5, 8\}$ . List the elements of: (i)  $(A \cap B) \times (B \setminus A)$  (ii)  $A \times B$

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AMBROSE ALLI UNIVERSITY, EKPOMA ;

we DIRECTORATE OF PART-TIME DEGREE AND DIPLOMA PROGRAMMES

DEPARTMENT OF COMPUTER SCIENCE LE

' First Semester Examination, 2012/2013 Academic Session

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| eae dae (Discrete Mathenratics)

ty,

tnstruction: Allemt any FOUR questions ° Time: 21/2 nrs. \*

Ql. (a) Let the universal set  $U = \{x \text{ is a positive integer, } 13 \leq x \leq 10\}$  . '

Cc

$A = \{1, 2, 3, 4, 5, 6\}$ ,  $B = \{2, 7, 8, 9, 10\}$ . Find (i)  $A \cup B$ , (ii)  $A \cap B$  (iii)  $A \setminus B$  . |

, (k) la survey of 40 people, iL was found that 28 read Newsweek magazine, 26 read Tell

| . Magazine, and 25 cad Newswatch. 9 also read both Newsweek and Newswatch, I read

: both Newsweek aid Tell, 8 read both Tell and Newswatch and § read no magazine at all.

(i) Find thenumber of people who read all three magazines. "ose Cc

(ii) Fill in Che number of people in the regions of the Venn diagram. - AU é {

(iii) Determine the humber of people who read exactly one Magazine. (2 Ei g f

. : i . : 'on [Oo . ~ ae

i Find A — 247-54 71. ee \ \_

' \_ (d) (i) Define mayping - . ; co ;

a | ; (ii) Given the lollowing mappings:  $f = (xy, 2)$ ,  $T = \{1, 2, 3\}$ , and  $U = \{a, b, c\}$  . i

\ and  $2 T \text{ by } f(x) = 3$ , Mylo si = % gi:  $T \rightarrow U \text{ by } g(i) = b$ ,  $g(2) = c$ ,  $g(3) = a$ . 2

(c)  $WA =$

soit

ae a t , a "

Pf). ane! e) "\ y d A

3 a2. fa)  $f(x)$  Courpute SS) and WW  $f(x)$  |

ned, xed

(v) Expand  $(2x + 3y)^n$  lp terms of x and y.

Hy (c) Use the principles-of mathenv! \* duction to slsow that

YaSe5k ee heoPn |

Find the: 4 tne recurrence relation  $a_{n+1} - 5a_n + 4a_{n-1} = 0$  . j

with the initial conditions  $a_0 = 1$ ,  $a_1 = -1$ . ' . 4. or

. " 4,

Find the generating function of the sequence |, a ca yoo,

where cris a fixed constant.

1 ab. Giventhat Aaege}d 2 (a) tvalueate  $[A]$ , the determinant of A.

2-2 I ,

(b) Find the adjoint of the above matrix A. Hence find  $A^{-1}$ , the inverse of A.

ny  $a \sim : . i$  ,  $ae \notin ig$ ” hee

See FH) Find (1) the characteristic polynomial of A - .

(ii) the eigenvalues of A

- (iii) the corresponding eigenvectors of A.  $\sim$  , 7

-OA- {i} (i) Define Spanning Tree of a connected graph , acne oy

\_ (i) Give the Greedy algorithm for a spanning tree , :

(iii) Define a minimal spanning tree - :  $\sim$

iv) Give the Prim’s algorithm for a minimal spanning tree

b) Given the post order and in order traversal of a binary tree, draw the unique binary tree. Postorder: dcblhpa , Inorder, — deebfahyi

(.)(.) Use a binary tree to represent the expression  $((a + b) * c) + (d * e)$

(i) What is the value of (i) prefix expression  $X44 \ 4G / 42 \ 823 * T03/4$

(d) (i) Give “the Breadth First Search (BFS) algorithm for the shortest path in a graph without weights ; ;

(ii) Find the shortest path from vertex s to t and its length in the graph given below

ub, (a) Define the following : ,

: (i) graph, (ii) directed graph, (iii) complete graph, (iv) strongly connected graph. :

(b) Give the following adjacency matrix- BFS

2 3% SL

0.000120 Up

4440 0 0.1, a

Nghe, 1 0-0 1.0 (i) Find  $S = [S,]$ , the symmetric matrix (G)

rk LO)

hy 0.0 0

(ii) Show that  $G$  has or does not have a clique. . \

(i) Define the following: , -

er path (ii) Euler circuit: (iii) Hamilton path (iv) Hamilton circuit.

“Use the Welsh- Powell algorithm for vertex coloring a graph

the Welsh- Powell algorithm to color graph G and find (9)

AMBROSE ALLI UNIVERSITY, EKPOWA

FACULTY OF NATURAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

First Semester Examination, 2013/2014 Academic Session

| CSC 309 ~ (Discrete Mathematics)

1 - Instruction: Attempt any FOUR questions. Time: 2 hours

\_ Q. (a). (i) Define the modulo operation. Find  $22 \bmod 9$  ,  $57 \bmod 12$ .

(ii) Define one-to-one and onto mappings. Give an example of each

{t "ME PR» Roane gs R—y Kare defined by the formulas: ;

| Payers Qasr', find fa) sg, fO)s Lec) s fo f(2).

4 ew

| \_ (b) HA = (1,3,5,7) and B = (2,3,5, 8) List the elements of

| (i) (A c BY X (B-A) (ii) (A- B) X (B-A) (iii) AX.

! | (c) Define the following and give an example in each case

(i) Normal fuzzy set (i) o. level eut, (i) Absolute and Relative complement of a fuzzy sul.

In a group of 79 cars tested by men of FRSC in Benin-City, 15 had faulty tyres, 20 had faulty brakes and 15 had no break lights. Also, 10 cars had faulty tyres and 4 cars were unsatisfactory in all three respects. How many cars had no faults in these checks? Draw an appropriate Venn diagram, uw

Q2 (a) ; Let R be the relation from A = {1,2,3,4} to B = {x, y, 2}

' : defined by R(1,y). (2) (3,y). (4,x), (4,2)

L (i) Determine the domain and range of R ;

F (ii) Find the inverse of R ; et

' \* (b) Define the following and give examples. , example of each Yio 2 4

i we (i) A group (ii) A subgroup (iii) A normal subgroup

(iv): Show that the set of non-singular 2 x 2 matrix is a group under multiplication

i (c) set = {1, 3} and } = e ; "a compute the Following

4 OeGg ae f

i 5 oe fy, a0, & WP 5 Bw", (fier) 1 (a py"

(ii) Write out the elements of a permutation group .s, .

VW («) ~ Write the following as a single cycle or product of disjoint cycles

| : (i) as a real tii) (1 2)(1 3 4), i) CA eo yea

i . Vo eB PL GAL 20 ye gee bey ape we eo -{"

OF Cay of IR)~ C152) i) yy is

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: — . a ae , , : x

ft La 4-8 or = 2 cad

zl 3. (a) Define the following. ; (i) path (ii) directed Graph (iii) planar graph

q \* (iv) State the Wetch- Powell algorithm for vertex coloring aap

. ; 0 Ol

(b) . Given A(G) = : m3 ; . (i) Draw the digraph, + fe

pee 110 .

é (li) Show whether the digraph have or do not have a clique.

(c) (i) Give the Breadth first search algorithm for the shortest path ii. "graph without weights

(ii) Use the algorithm to find the shortest path from vertex s to vertex t and its length in the given graph

(i) What is coding theory? Give a schematic diagram of the steps in (range finding

information. Use the 7-bit Hamming code for the following :

(ii) Explain the process of encoding and decoding :

(iii). Encode the following message into code word 1104

(iv) Decode the following received transmissions 9011001

ae : oO mek opera Sure ~

b l cee) TC @ipd={20 1 3) Find A? SA 467 (MA S31 >| A

ee re ed ae) ce AN 2 ly

(c) Write the matrices for the following Quadratic form 'yey

'| \_ — ; : (i} aa 45x32 45x23 — = 2x,% + 2x), + 6x5X, 0° a Ma \

(ii).  $8 \sim 2x) - Tx\} - 4x,x, + 8x,x, : Sh$

| Write the Quadratic form corresponding to the matrices a \ i & Ly q

'h 2-34 2 1 -2

| eo (ii) |-3 200 47 (iv) |} 1 2 -2 aes

l 8 n

! Q5. <a) Compute > FQ) and FL fy)

(b) Expand  $(2x + 3y)^5$  in terms of x and y..

(c) (i)-Use the principles of mathematical induction to show that

$7 : 1+3+5+ \dots + 2n-d = n^2$  '

| (ii) Find the solution of the recurrence relation  $a_{n+1} - Sa_{n+1} + 6a_n = 0$

"|; with the initial conditions  $a_1 = 1, a_2 = 1$ .

i . (ii) Find the generating function of the sequence  $1, a, a^2, \dots$ ,

where «is a fixed constant.

AMBROSE ALLI UNIVERSITY, EKPOMA

DEPARTMENT OF COMPUTER SCIENCE

DIRECTORATE OF PART-TIME DEGREE PROGRAMME

2014/2015 FIRST SEMESTER EXAMINATION

. & "CSC 309- DISCRETE MATHEMATICS

INSTRUCTION: Answer question ONE and any other three. ' Time: 2hrs. 15mins |

\* Q1a (i). Define discrete mathematics |

(ii) List at least three basic discrete mathematics topics

b(i). Define a countable set

t/t Us 1,2.08%  $A = \{1,2,3,4\}$ ,  $B = \{2,4,6,8\}$  and  $C = \{3,4,5,6\}$

Find (i) AS (i) (ANC)\* (iii) BYA . oe

(iii) Represent the following set expressions using Venn diagram: (i)  $B^c$  (ii)  $(B \cap A)^c$

(iv) In a school, 220 students offer Computer Science or Mathematics or both. 425 offer Physics and 110 offer Mathematics. How many students offer Computer Science but not Mathematics?.

C. Define (i) Tautology (ii) a truth table

(iii) Find the truth table of the following statements: (a)  $\sim (p \sim q)$  (b)  $(p \vee q) \wedge (p \wedge q)$ .

Q2a. Define the following: (i) a graph (ii) a complete graph: (iii) a digraph

Give examples of each:

Distinguish between a graph and a digraph.

(i) Write out the vertices of  $Z$

(ii) Write out the edges.

b. Consider the digraph given below:

c(i) Let  $A = \{a, b\}$ ,  $B = \{1, 2, 3\}$  and  $C = \{x, y\}$ , find  $A \times B \times C$

(ii) let  $A = \{a, b\}$ ,  $B = \{2, 3\}$  and  $C = \{3, 4\}$ , Find (a)  $A \times (B \cup C)$  (b)  $(A \times B) \cup (A \times C)$

Qe = vs, erry,

a = ( car, ' )

° : : i \_ An C = +3 8

aS Define (i) a fuzzy set (ii) a normal fuzzy set (iii) a union of fuzzy set

Q3a. Define (i) a formal language (ii) a grammar (iii) a word

Why are grammars in formal language useful? |

b(i) Define a switch

(ii) Make a schematic diagram of the switching network ;  $(p \wedge q) \vee (p \wedge r)$

Given the real numbers 6, 2 and 0 with membership grades 0.2, 0.8 and 1:

Write the fuzzy-set.

Q4a. If  $A = \{1, 3, 5, 7\}$  and  $B = \{2, 3, 5, 8\}$ . List the elements of : (i)  $(A \cap B) \times (B \setminus A)$  (ii)  $A \times B$

c :

b(i). If  $A = (5 \times)$  Find  $A \cap 3A - 134$

L

(i)  $\text{tr}(Q)$ , Find  $a$ ?

(iii) Is the matrix  $A$  in d(ii) above non-singular? Give reason for your answer. | 1 ude

Qa (i) Define one-to one and onto mappings.

ti) if  $f: R \rightarrow R$  and  $g: R \rightarrow R$  are defined by the formulas:

$f(x) = x^4z$ , at  $x=?$  find  $f \circ g(x)$ ,  $f \circ g(-2)$ ,  $f \circ g(2)$

b. The following diagram defines functions  $f, g$  and  $h$  which map the set  $\{1, 2, 3, 4\}$  into itself.

{i) Find the range of  $F$ ,  $g$ , and  $h$  : oh ta

(ii) Find the composition functions (a)  $f \circ g$  (b)  $h \circ f$ .

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AMBROSE ALLI UNIVERSITY, EKPOMA

'FACULTY OF NATURAL SCIENCES

"DEPARTMENT OF COMPUTER SCIENCE

First Semester examination, 2011/2012 Academic Session Pp ~

CSC 309 (Discrete Mathematics) 2

Instruction: Attempt any FOUR questions in 2h

Time: 2 1/2 hrs.

Q1. (a) Let the universal set  $U = \{x | x \text{ is a positive integer, } 1 \leq x < 10\}$   $A = \{2, 3, 4, 5, 6\}$ ,  $B =$

era rind (i) AWB, (i) (AB) (ii) AS BS

-(b) IFA, = (4, 23), = (3,405), Ay = (3,6,7), Aq = {3,8,9) and | = {2,3,4). Find (i) ai (ii) F - ind

(c) FB = (1,4); C= {4,5}, and 0 = (5,7). Find (8x C) U (Bx D), (ii) (B xC) A(B xD). (iii) (B Ww D)x(D—C). 7

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2g)

(d) fA = |; || MR Be 0

ae)

Des AI, 223 , 6). Define a ele Ron A by writing (x y)ER

“(b) ss Consider thesete

ifand only.if y- isa eat of 3 ere, iia

ae (i) Describe'R as a subset of AXA ; Reece:

(ii) Show that R is an equivalence relation on A.

(c) (i) Find all integers x such that - 95 <x <25 and  $x = 3 \pmod{5}$ .

(ii) Find allx such that  $0 < x < 6$  and  $2x = 4 \pmod{6}$ . .

a2. (a) Define the following |

(i). Group. (ii). Subgroup {iii), Normal Subgroup,

(b) (i) State the elements of a permutation group Sy

(ii) Give the Cayley table for this group.

(cd Show that the lower triangular, matrix efthe form  $f = \begin{pmatrix} & \\ & \end{pmatrix}$  where ad #0

. c 6

isa subgroup but not a normal subgroup.

» -Q3.fa) (i) Expand  $(v - 3y)^*$  in powers of x and y.

y-2 4 6

(ii) Solve the equation  $| | \phi + 1./-7| = 0$

0) 0 1-4

wate Yb yprz—2w=-4

(b) Solve.the linear system ay at i

. 2ynd yr-st2w=5

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s : \ ,

(c) bet a at ; ; 4 and fh = ( ; ; y } 'Compute the following

O fow (i) eof Gi) a' of (iv) fea' (vy) (Boa) (vi) (ao py

<< Q4 (a) Given the Boolean expression  $(avbveja(a'vbveja(a'vbvc'y$

Design a circuit corresponding to the expression:

(i) Lattice: - ,

{ii} Partially Ordered Sets { #

(iii) Show that the relation  $\leq$  is a partial ordering on the set of integers.

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0,4 1 0 2.0

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(c) Sketch the graph of the following matrix

Qs. a6) : Define the following (i) Directed graph (ii) Planar graph (iii) Path \* Rooted tree

(b) What is the difference between the matrix of a graph and its adjacency matrix?

| ; (c) — . Determine whether the digraph with given adjacency matrix is strongly connected?

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4(0 7 lo.

y 8130 0 1 1 6 sacle cl '

is . at > the 7 . + :

clio 04 determine the Clique in Y, if any

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Instruction:

Qt. (a).

(b)

(0)

Q2 (a)

—(b)

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AMBROSE ALLI UNIVERSITY, EKPOMA

FACULTY OF NATURAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

First Semester Examination, 201 3/204 Agattomiic Session

Digeret Mathematic. )

\_ Alleni; (any FOUR questions 8 i ' Time: 2 4/4 hrs.

( i) Defining the mod function, Find  $22 \bmod 9$  ,  $57 \bmod 42$ ,

(ii) Define one-to-one and onto mappings, Give an example of each.

“(i) If  $f, g: R \rightarrow R$  are defined by the formulas :

$f(x) = 2x + 4$   $g(x) = x^2$  find  $f \circ g$  ,  $g \circ f$   $f \circ f$   $g \circ g$  :

$A = \{1,3,5,7\}$  and  $B = \{2,3,5,8\}$  1. Give elements of :

$(A \setminus B) \cap (B \setminus A)$  (ii)  $(A \cap B) \cup (A \cup B)$  (iii)  $A \times B$  z

| Define the following and give an example in each case where ( )  
 i Normal fuzzy set ( ) level cut (ili): Absolute and alti complete of a fuzzy set &  
 in a group of 76 cars tested by men of FRSC in Bbnin- city, 15 had faulty tyres, 20 had  
 ' faulty brakes and 18 ' had broken lights, Also, 5 had faulty tyres and breaks, 6 failed  
 on tyres and brake- lights and 4 cars were unsatisfactory in all three respects. How many,"

pete had no faults in these checks? Draw a pre Venn diagram, "

Let R be the relation from  $A = \{1, 2, 3, 4\}$  to  $B = \{y_1, y_2, y_3, y_4\}$  defined by  $R = \{(1, y_1), (2, y_2), (3, y_3), (4, y_4)\}$  is a function

(i) Determine the domain and range of R respectively

"ii Find the inverse of R

: \(\mathbb{Z}\) to \(\mathbb{Z}\)

Define the following and give an example of each halom if 7 yO,

(i) A Group (ii) A subgroup (iii) A normal subgroup

(iv) Show that the set: G of non singular  $2 \times 2$  matrices is a group, under multiplication  
 ! 312

(il) write out the elements of a group of order 4

» (i) let  $A = \{1, 2\}$  and  $B = \{x, y, z\}$  then loge

Write the following as a single cycle and disjoint cycles: (1 2 3 4 5 6 7 8 9)

12273 45 6 f :

ut  $4 \times 5 = 20$  (ii)  $(4 \times 5) \times (2 \times 3 \times 5) = 4 \times 5 \times 2 \times 3 \times 5$

SBP by dorg—eo- A tile 5 4 be yet\ ytae

hl bog cles Cig sas Ve 5 uf

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— Q3.(a) \* — Define the following (i) path : (ii) fitted path (iii) "planar graph

" (iv) State the edge coloring for vertex coloring in a

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(b). Given  $AIG = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100\}$  Braye bara, tia

uP EEE Lo: Te ee

ee ' get | (ii) show whether the digraph has or does not have a unique

; (c) + (i) Give the Breadth first search algorithm for the shortest path in a graph without weights

ey (i) Use the algorithm to find the shortest path from Vertex 5 to vertex 10 and its length in

> the given graph' Haas

"at (i) What is coding theory? Give a schematic diagram of the: is a code in a graph

information: Use the 7-bit Hamming code for the following.

i Explain the process of encoding and decoding with an example

(iii) Encode the following: Message into code word Hot a

(iv). eode pope fecelved Kransmissions 0011001 Pelee

Fey te ies | Bt a Find RSH +61 Wi. IA a3

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i | ;

(yes Write the matrices for tha folowing Quadra for

(i  $3x + 5x? + 5xy = = 2xjx, + 2x xy + 6x 3)$

i

' x) af = i200} ~ ce 4x, x, + 8x, x, |

'Write Ve" aeale form corresponding to the matrices

2. 3 pei a

Gi) 4] £3 a PAM EPL 2D: bol

5 Lf 4 [e2s-at

Q5, (a) ee | — bi Compute > 'ay we

'() | Expand  $(2x + 3y)^6$  in terms of x andy. Hedy Hp

lc ) (i): 'Uso 'the principles of mathematical induction to show that i

ae  $143H5 + i. 2 4 nef = ne a \}$  a

(li) Find the solution of the 'Tecurrence rg lation ab eas: Say at 6a, +10

with the inital conditions a," =11, 4 ES 7h ' me

iii) Find the generalih function of the sequencel 1, a, a? sores a

\_ where ais a fited constant, ere

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FACULTY OF NATURAL SCIENCES

\_ DEPARTMENT OF COMPUTER SCIENCE TR re:

pe fa "First Semester Examination, 2014/2015, Academic Session 7

CSC 309 (Discrete Mathematics)

instruction: \_ 'Attempt any FOUR questions ' Time: 2h

; Ved ime: 2hrs.15mins.

Q1. \~ (a) (i) Define the mod function. Find  $22 \bmod 9$  ,  $57 \bmod 12$ .

(ii) State and explatn the set of numbers

BB a)  $P:R \rightarrow R$  and  $gk \rightarrow R$  are defined fie formulas:  $f= x^2 \log(x) = 3?$

find  $f.e(x) . gy fC), L8(-2) . Bf).$

- i <= (iv) Let  $A_n = \{ie Z: \text{is-divisible by } n, \text{ sacri Find Ay Ar MOA.}$

ra  $\{b\} HA' = 5,7)$  and  $B = \{2,3,5,8\} C=(1,2,4, |$  Listthe elements of |

() nay A) (i)  $(A-8)X(B-A)$  (ill)  $AX(BLUC), |$

ate owe 2 1 -3| Find 42 -SA + Or (IPAS [3 Ibo 2 'Find 47 Is

[yt 10 OA Es

4 : - 7 (

\* (d) In a class of 15 Computer Science students, 63 have taken calculus but not chemistry. If the number of students who have taken calculus and chemistry is 4

i.  $\sim$  chemistry | pe nat fall. : | te \

Let  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ , and let  $R$  be the relation on  $A$  defined by  $xRy$  if and only if  $x$  is a multiple of  $y$ .

(i) Write down a set of ordered pairs for  $R$ . (ii) Determine the domain of  $R$ .

(iii) Find the inverse  $R^{-1}$  of the relation. (iv) Find the range of  $R$ .

(v) Find the composition  $R \circ R$ .

(vi) Find the transitive closure of  $R$ .

(vii) Find the reflexive closure of  $R$ .

(viii) Find the symmetric closure of  $R$ .

(ix) Let  $R$  be the relation on the set  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  defined by  $xRy$  if and only if  $x$  is a multiple of  $y$ .

(x) Find the range of  $R$ .

(xi) Find the domain of  $R$  and write down a set of ordered pairs for  $R$ .

(xii) Define a relation  $R$  on the set  $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$  by  $xRy$  if and only if  $x$  is a multiple of  $y$ .

(xiii) Show that the set  $G$  of non-singular  $2 \times 2$  matrices is a group under matrix multiplication.

(xiv) Show that the group  $(\mathbb{Z}/n\mathbb{Z}, +)$  is isomorphic to  $(\mathbb{Z}/n\mathbb{Z}, +)$ .

(xv) Let  $A = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$  and let  $R$  be the relation on  $A$  defined by  $R = \{(a, b), (b, c), (c, d), (d, e), (e, f), (f, g), (g, h), (h, i), (i, j), (j, k), (k, l), (l, m), (m, n), (n, o), (o, p), (p, q), (q, r), (r, s), (s, t), (t, u), (u, v), (v, w), (w, x), (x, y), (y, z)\}$ .

(xvi) Find the transitive closure of  $R$ .

(xvii) Find the reflexive closure of  $R$ .

(xviii) Find the symmetric closure of  $R$ .

(xix) Let  $G$  be a graph with vertex set  $V(G) = \{a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z\}$  and edge set  $E(G) = \{(a, b), (b, c), (c, d), (d, e), (e, f), (f, g), (g, h), (h, i), (i, j), (j, k), (k, l), (l, m), (m, n), (n, o), (o, p), (p, q), (q, r), (r, s), (s, t), (t, u), (u, v), (v, w), (w, x), (x, y), (y, z)\}$ .

(xx) Find the chromatic number of  $G$ .

(xxi) Find the clique number of  $G$ .

(xxii) Find the independence number of  $G$ .

(xxiii) Find the diameter of  $G$ .

(xxiv) Find the radius of  $G$ .

(xxv) Find the girth of  $G$ .

(xxvi) Find the degree of  $G$ .

(xxvii) Find the order of  $G$ .

(xxviii) Find the size of  $G$ .

(xxix) Find the number of vertices of  $G$ .

(xxx) Show that the graph with adjacency matrix  $A(G) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$  is a cycle graph.

(xxxi) Find the eigenvalues of  $A(G)$ .

(xxxii) Find the eigenvectors of  $A(G)$ .

(xxxiii) Find the characteristic polynomial of  $A(G)$ .

(xxxiv) State the Dijkstra's Shortest Path algorithm.

(b) Use Dijkstra's algorithm to find the shortest path between node 1 and every other node in the following

~~ simplified Reinert, pee : |

ie the matrices for the following Quadratic form  $Q(x,y,z) = 4x^2 + 5y^2 + 3z^2 + 2xy + 4xz + 6yz$  | a ore

(i)  $B = \begin{bmatrix} 4 & 2 & 4 \\ 2 & 5 & 6 \\ 4 & 6 & 3 \end{bmatrix}$  (ii).  $P = \begin{bmatrix} 2 & 4 & 6 \\ 4 & 5 & 3 \\ 6 & 3 & 4 \end{bmatrix}$  Ax

E Wie the Quadratic form corresponding to the matrices). c pA i .

(ii) Yeh) 22 Pe) ee See yo 2

eA 42) 3 e re)

: (b) oe is ae ne Compuc ue f(x) ie 5

sequent , a, at i: Set "whet ee pt constant.

(a) (i): State the pn of Dusty of 8 Bodleian agebia, (i) Given that ' a beoben algebra and i

+1 110. alate O. : : reas .

i} al i fod ae 0 eta = 1101010.ans be so1013 in. Using the e,find "a+b, a\*b.

(b) Given the Boolean expression  $(A \oplus B) \vee (C \wedge D)$  | Aba oh Mv te nb A Ce

(i) Design a circuit corresponding to the expression. (i) Simplify the € expression.

| (i) Design the equivalent clrcuit from the simplified Gepressm ; :

———s

Lax: tdxxy Bxt-t [Ke o2

ADMBROSE ALLE UNIVERSTEY, EKPOMA K

DEPARTMENT OF COMP VEER SE TENG 0

DIRECTORATE OW PARTSTIME: DEGRIN PROGR. AXYMES

2013-2014 FIRST SEMESTER DE GREL XAMIN, ATION

CSC 309- DISCRETE MATIE MATICS

CRE TTONS: Answer question ON}! and anyother fliree questions. Time: 2hrs

Qla. Define (i)ja set (ii) a countable set (iit) cardinality ofa finite set. (J mk cach)

(iv) State the conditions fora set S to be countable. (2mks)

(v) Given. that  $A = \{x \in \mathbb{N} : x \text{ is odd}\}$ ; Does  $A \cap \mathbb{N} = \emptyset$ . (Sms)

(vi) Let  $A = \{1, 2, 3\}$  and  $B = \{a, b\}$ , find  $A \times B$ . pote)

bh. Let  $A = \{123, 456\}$ , Bo "(2.7.89 40), Ue  $\{4, 2, 3, 10\}$ ,

yoorls

find () Aups (ii)  $(A \cap B)$ . (2mks cach) :

lus th, 248s J and '

c. Let U the univ ergal set be the set of positive integers;

let  $A = \{1, 2, 3, 4, 5\}$ ,  $B = \{2, 3, 4, 7\}$ . no |

yy find (i)  $A \cup B$  (ai)  $A \cap B$  (iti)  $A \setminus B$  (iy)  $A \times B$  (Zmiks)

d. Use veun diagrams to verity the validity of ihe following ar guments.

(i) the RAM i ix in (he CPU

(ii) No girl is a wom: in

(iii) All boys)

(iy) Sonre WW ma}

f

Q2h. Define § asca ft: ay Ard X 'Gi) a sy mimetric matrix. (Znaks cachy i

" Give example pha scalar matrix ef order 3.(rmik) | i

h(i) Whenjare oy mate ices Aand B said to be conformable for multiplication?.

-(2miks) |

mri ane ile ee

Show thatA ake iB are eh a able for the product A

JA sample of 80 car owners revealed fhal 24 ow ng } §

which re. tot jeeps. Wind the Auber of people who; owned both i jeep: and a some other ae cae) ce |

aie fen (i)a juny set (i) a nornial Vuvzey set (in) union of Muzzy sets. (2mks

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buG viven (he real numbers 3, band 0 with member sip gra ades

weal he Vurzy. set. (mils) |

©. A= (1,4), B= (4,5),C= 6, 1), Cire () [AN VUUAXC ') "i) (A v C) X(C-A) |

ci AX«B Ua (2mks packs. | a Su oO +62 \_ae

¥ C1509 and 1.0.

if i 2 [debe Souk aay

Qda. Defi | :

a. Define (i) a mapping from a set S toa set T. (ii) a oe i i

> « a a relation le ase

(iii). Domain of a relation R. (2mks Bach oma set A to B

Pane the following diagr am define functions f aaa g 8 which map, the Set (1,2,3,4) into

: gel, eg |

(i) find the vange Cort and g. (3mks) beg ay ae conn

(il) find the ¢ omposition functions of fo 2. (Jaks) i

(iii) Under y what condition will a function haye an iene' (imk) i

ue If A= hy 2,5), B=-(2 4), find AXB.(Imk) bg re Po Pos: f

d. Given Ax (234) and B= GaSe Pole Bete

' 10) list all ck' ments 'of cach relation Ro 'delined above. (2mks), : IK

: (ii) list (he) domain and range. Camis: | ' Ro i :

\ hon Define { (ia geaph (ii) a complete graph iii a bier ante Giles : |

is oe vA footall {ournament consists of Steams: 11, 12,13 "4 an foe

dines sik sites \_ Assyme that latter al number of pames he avg been played, he h ve aye folld ving |

a has played 2,13, Fdyand S) | | i ae ea be \

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has pla 1, v2 and \4

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pepe escv! t nv Whove situation byt a graph and: wr ite: ut the vit ces. (7miks) 1

ag @)y-Com ate “the le of the mate ix: (mls)

(uy Is the matrix A inn singular?. (fang Give reaso 1 for ‘Your 4

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AMBROSE ALLI UNIVERSITY, EKPOMA |

FACULTY OF PHYSICAL SCIENCES

DEPARTMENT OF COMPUTER SCIENCE

2016/2017 FIRST SEMESTER B.Sc DEGREE EXAMINATION

COURSE CODE/ TITLE: CSC 309- DISCRETE MATHEMATICS

INSTRUCTION: Answer question 1 and other three questions. Time: 2hrs 30 mins

Q1a. Define the following: (i) directed graph (ii) planar graph (iii) Hamiltonian epee ero eenl

b(i). When do we say that two graphs are isomorphic?.(2mks) ~ Nero 09K WHESPOMC

ji). State the Euler’s formula for a connected planar graph. (1mk) —

\pg ut <2

c. Consider the adjacency matrix below: p

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f- grt

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Ril 01000 f=:

(1) Draw the graph representing the adjacency matrix.(3mks)

(II) Find  $C = A(G) + [A(G)]^2$ ? (6mks)

(III) Is the resultant adjacency matrix strongly connected?. (1mk)

Give reason for your answer.

7 '

Q2a. Define the following (i) a graph (ii) a complete graph (iii) an adjacency list (iv) an adjacency matrix.

(2mks each) SI0ro

b. Represent the undirected graph below by an adjacency linked list.(4mks)

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c. Write the adjacency matrix of the digraph below: (5mks) ec) © he com)! gen?

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a i, P,

Q3a. Define the following: (i) a vertex (ii) a path (iii) an edge (iv) a loop. (1mk each) >

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oyo3e?

b. The figure below shows the graph of Nigeria:

Nigeria

Bayelsa

Rivers

ae

Rumukoro PH City Trans- Otuoke Ekpoma

Bouses Amadi

(i) vertices (ii) path (iii) edge (iv) degree (v) the root node.

Yenagoa Asaba

Identify the following in the figure above:

\*(1mk each) :

c(i) Draw a digraph determined by the given matrix: (3mks)

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1001

110 0

0 1 0 0

d. Find the solution of the recurrence relation:  $a_n - 5a_{n-1} + 6a_{n-2} = 0$ , with initial conditions  $a_0 = 1$ ,  $a_1 = -1$  -

(Smks)

Q4a. Define the following: (i) one —to- one mapping (ii) onto. mapping (1mk each)

b. If  $f: R \rightarrow R$  and  $g: R \rightarrow R$  and defined by the formulas :

$f(x) = x+2$ ,  $g(x) = x$

\_find (i)  $f(g(x))$  (ii)  $g(f(x))$  (iii)  $f(g(-2))$  (iv)  $g(f(2))$ . (8mks)

c. The digraphs below define functions  $f$ ,  $g$  and  $h$  which map the set  $\{1,2,3,4\}$  into itself.

$f$  is  $\&$ ,  $h$

'4' . ; : 1

2 2 2 2

3 3 3 3

ae 4

(i) Find the range of  $f$ ,  $g$  and  $h$  (3mks)

(ii) Find the composition functions: (a)  $f \circ g$  (b)  $h \circ f$ . (4mks)

Q5a.  $f(x) = (x+1)/x$ , compute  $\bigcap_{x=2}^8 f(x)$ ,  $2 < x < 8$  (2mks)

b. Expand  $(2x + 3y)^n$  in terms of  $x$  and  $y$ . (5mks)

c. Simplify (i)  $(n-1)!/(n+2)!$  (2mks) (ii)  $1/e$ . (2mks)

kel

d. Find  $n$  if  $2P(n,2)+50 = P(2n,2)$  (6mks)